

Week 1 Task: Retail Sales Data Cleaning & Basic Analysis in Excel

Tool: Microsoft Excel	Duration: 7 Days
Dataset: retail_sales_dataset.csv	

[Click to download Dataset](#)

Objective

The objective of this task is to help you understand how to work with a real dataset in Excel. In this week, you will learn how to open a dataset, understand its columns, clean and format data, apply basic formulas, use sorting and filtering, and write simple observations.

This is a beginner-friendly task. The main goal is not to create an advanced dashboard, but to understand how raw data is prepared for analysis.

Dataset Overview

You are given a retail sales dataset. The dataset contains sales transaction records from a retail business.

The dataset includes the following columns:

Column Name	Description
Transaction ID	Unique ID for each transaction
Date	Date of the transaction
Customer ID	Unique ID of the customer
Gender	Gender of the customer
Age	Age of the customer
Product Category	Category of product purchased
Quantity	Number of items purchased
Price per Unit	Price of one unit
Total Amount	Total sales amount of the transaction

Your Task

Part 1: Open and Understand the Dataset

Open the dataset in Microsoft Excel and answer the following questions in a separate sheet named **Dataset Understanding**:

1. How many rows are in the dataset?
2. How many columns are in the dataset?
3. What type of data is stored in the **Date** column?
4. What type of data is stored in the **Product Category** column?
5. What type of data is stored in the **Total Amount** column?
6. What are the unique product categories available in the dataset?
7. What is the minimum and maximum age of customers?

Part 2: Format the Dataset Properly

Create a copy of the original dataset sheet and rename it:

Cleaned Data

In this sheet, perform the following formatting tasks:

1. Convert the data range into an Excel Table.
2. Make the column headers bold.
3. Adjust column width so all data is clearly visible.
4. Format the **Date** column as a proper date.
5. Format **Price per Unit** and **Total Amount** as currency or number format.
6. Make sure all column names are clean and readable.
7. Freeze the top row so headers stay visible while scrolling.

Part 3: Data Quality Check

In a new sheet named **Data Quality Check**, check the following:

1. Are there any blank cells in the dataset?
2. Are there any duplicate rows?
3. Are all values in the **Quantity** column valid?
4. Are all values in the **Price per Unit** column valid?
5. Does **Total Amount** match this formula?

`Quantity * Price per Unit`

Create a new column in the **Cleaned Data** sheet named:

Calculated Total

Use this formula:

`=Quantity * Price per Unit`

Then compare **Calculated Total** with **Total Amount**.

Create another column named:

Amount Check

If both values are equal, show **Correct**. Otherwise, show **Check**.

Example formula:

`=IF([@[Calculated Total]]=[@[Total Amount]], "Correct", "Check")`

Part 4: Basic Excel Formulas

Create a new sheet named **Basic Analysis**.

Use Excel formulas to calculate the following:

1. Total sales amount
2. Average sales amount
3. Minimum sales amount
4. Maximum sales amount
5. Total quantity sold
6. Average customer age
7. Youngest customer age
8. Oldest customer age
9. Total number of transactions
10. Total number of unique product categories

Use formulas such as:

`=SUM()`

`=AVERAGE()`

`=MIN()`

`=MAX()`

`=COUNT()`

Part 5: Sorting and Filtering

In the **Cleaned Data** sheet, apply sorting and filtering to answer these questions:

1. Which transaction has the highest total amount?
2. Which transaction has the lowest total amount?
3. Show only transactions where **Product Category** is **Clothing**.
4. Show only transactions where **Gender** is **Female**.
5. Show only transactions where **Quantity** is greater than 2.
6. Sort the dataset from highest **Total Amount** to lowest **Total Amount**.
7. Sort the dataset from youngest customer to oldest customer.

Take screenshots of at least 3 filtered or sorted results and paste them into a sheet named:

Sorting and Filtering Screenshots

Part 6: Simple Category-Based Analysis

In the **Basic Analysis** sheet, create small summary tables for the following:

Sales by Product Category

Create a table like this:

Product Category Total Sales

Beauty

Clothing

Electronics

You can use `SUMIF` for this.

Example:

```
=SUMIF(Product_Category_Range,"Beauty",Total_Amount_Range)
```

Quantity Sold by Product Category

Create another table:

Product Category Total Quantity Sold

Beauty

Product Category Total Quantity Sold

Clothing

Electronics

Sales by Gender

Create this table:

Gender Total Sales

Male

Female

Part 7: Basic ChartsCreate a new sheet named **Charts**.

Create the following charts:

1. Bar chart: Total Sales by Product Category
2. Column chart: Total Quantity Sold by Product Category
3. Pie chart: Total Sales by Gender

Each chart must have:

1. Clear chart title
2. Proper labels
3. Clean design
4. No unnecessary colors or extra effects

Part 8: Write Simple ObservationsCreate a new sheet named **Observations**.

Write at least 5 simple observations from your analysis.

Your observations should be based on the data, not guesses.

Example observations:

1. The highest sales category is _____.
2. The lowest sales category is _____.

3. Female customers generated _____ total sales.
4. Male customers generated _____ total sales.
5. The average customer age is _____.
6. The total quantity sold is _____.
7. The highest transaction amount is _____.

Part 9: Upload Your Work on GitHub

After completing the Excel task, you must upload your work to GitHub.

GitHub Repository Name

Create a new GitHub repository with this name:

week-1-retail-sales-excel-analysis

Files to Upload

Your GitHub repository should include:

1. Completed Excel file
2. Dataset file
3. Dashboard or chart screenshots
4. README.md file

Final Submission Requirements

You must submit one Excel file with the following sheets:

1. Original Data
2. Cleaned Data
3. Dataset Understanding
4. Data Quality Check
5. Basic Analysis
6. Sorting and Filtering Screenshots
7. Charts
8. Observations

Important Instructions

1. Your GitHub repository must be public.
2. Your README file must be properly written.
3. Your LinkedIn post must include your GitHub project link and must mention LogicStack.
4. Do not upload incomplete work.
5. Do not copy another intern's README or LinkedIn post.
6. Use your own words in observations.
7. Keep your GitHub repository clean and organized.

Bonus Task

This part is optional.

Create one extra column named **Age Group**.

Use this logic:

Age	Age Group
18 to 25	Young Adult
26 to 40	Adult
41 to 60	Senior Adult
Above 60	Older Customer

Then find total sales by age group.

